

ANIMESH SAH

Ph.D. Candidate in Physics · Tata Institute of Fundamental Research, Mumbai, India

✉ animesh.sah@tifr.res.in  LinkedIn  Shin107  0000-0001-9541-916X

RESEARCH INTERESTS

Model-independent tests of the standard cosmological model; the impact of cosmic anisotropies, inhomogeneities, and bulk flows on cosmological inference; peculiar velocities; and the application of machine learning methods to large-scale structure analysis.

EDUCATION

Ph.D., Physics 2023 – Present (expected early 2027)
Tata Institute of Fundamental Research (TIFR), Mumbai, India | Advisor: Dr. Mohamed Rameez.

M.Sc., Physics 2020 – 2023
Tata Institute of Fundamental Research (TIFR), Mumbai, India | Integrated Ph.D. programme.

B.Sc. (Honours), Physics 2017 – 2020
University of Delhi, India

JOURNAL PUBLICATIONS

- [J1] **A. Sah**, M. Rameez, and S. Sarkar, “Pantheon+ supernovae corrected for progenitor age indicate the universe is decelerating,” *Mon. Not. R. Astron. Soc.* **549**, stag844 (2026) [arXiv:2606.09650].
- [J2] **A. Sah**, M. Rameez, S. Sarkar, and C. G. Tsagas, “Anisotropy in Pantheon+ supernovae,” *Eur. Phys. J. C* **85**, 596 (2025) [arXiv:2411.10838].

CONFERENCE PROCEEDINGS

- [C1] **A. Sah**, “Examining the cosmic anisotropy with the Pantheon+ supernovae,” in *Proceedings of the XXVI DAE-BRNS High Energy Physics Symposium 2024*, Springer Proc. Phys. **937**, 284–287 (2025).

RESEARCH EXPERIENCE

Anisotropy and progenitor-age bias in the Pantheon+ Supernovae Compilation 2023 – 2026
with Dr. Mohamed Rameez, TIFR; Prof. Subir Sarkar, University of Oxford; and Prof. Christos Tsagas, Aristotle University of Thessaloniki

- Detected a dipolar anisotropy in the Hubble expansion rate of the Pantheon+ Type Ia supernova catalogue exceeding 1.5 km/s/Mpc – larger than the claimed 1% uncertainty of the SH0ES measurement – with the deceleration parameter showing dipolar modulation at $> 5\sigma$ significance, using Maximum Likelihood Estimation across heliocentric, CMB, and Local Group frames.
- Showed that correcting for Type Ia supernova progenitor-age evolution shifts the monopole deceleration parameter to positive values, indicating that the isotropic expansion of the universe is decelerating rather than accelerating.

Tests of Tilted Cosmology 2025 – Present
with Dr. Mohamed Rameez, TIFR

- Leading development of a model-independent pipeline to test tilted Λ CDM cosmologies against upcoming LSST supernova data, validated using realistic simulations from SNANA.

Mock Catalogs for the DESI Fundamental Plane Sample 2025 – Present
with Dr. Shadab Alam, TIFR

- Building realistic mock catalogs for a photometric sample of $\sim 480K$ early-type galaxies ($z < 0.15$) from the DESI Legacy Imaging Survey, using the AbacusSummit N-body simulation suite as the underlying dark-matter field.

TALKS

The Concurrence of Mega-Surveys August 2026
International Centre for Theoretical Sciences (ICTS), Bengaluru | Flash talk (programme participant; see Schools & Workshops).

Supernova Cosmology: New Challenges and Opportunities June 2026
Yonsei University, Seoul | Contributed talk on Pantheon+ anisotropy and progenitor-age systematics.

Cosmic Lighthouses: Astrophysical and Cosmological Challenges with Type Ia Supernovae July 2025

Kavli Institute for Cosmology, Cambridge | Contributed talk on Pantheon+ anisotropy.

5th EuCAPT Annual Symposium May 2025
Online | Talk on anisotropy in Pantheon+ supernovae.

Pune-Mumbai Cosmology and Astro-Particle Meeting – Chapter 4 February 2025
Indian Institute of Technology, Bombay | Contributed talk on anisotropy in Pantheon+.

XXVI DAE-BRNS High Energy Physics Symposium 2024 December 2024
Banaras Hindu University, Varanasi | Contributed talk on cosmic anisotropy with Pantheon+.

A Multipolar Universe? September 2023
Aristotle University of Thessaloniki, Greece | **Invited talk** on Pantheon+ anisotropy.

Less Travelled Path to the Dark Universe March 2023
International Centre for Theoretical Sciences (ICTS), Bengaluru | Talk on cosmological analysis with the Pantheon+ compilation.

SCHOOLS & WORKSHOPS

The Concurrence of Mega-Surveys July – August 2026
International Centre for Theoretical Sciences (ICTS), Bengaluru | Project: reimplemented and trained a transformer-based decoder (GOTHAM) to accelerate halo catalogue generation from particle-mesh simulations via distributed multi-GPU training. Also presented a flash talk (see Talks).

Rethinking Cosmology December 2025
Raman Research Institute, Bengaluru

Pune-Mumbai Cosmology and Astro-Particle Meeting 2024 – 2026
TIFR, Mumbai; IISER, Pune | Chapters 3, 5, and 7. Chapter 4 talk listed under Talks.

Largest Cosmological Surveys and Big Data Science May 2023
International Centre for Theoretical Sciences (ICTS), Bengaluru | Project: reanalyzed Dark Energy Survey cosmic-shear measurements using machine-learning techniques.

TECHNICAL SKILLS

Statistical methods	Frequentist and Bayesian inference, Maximum Likelihood Estimation, MCMC, Hamiltonian Monte Carlo (HMC) (emcee, NumPyro).
High-performance computing	Job submission, scheduling, and parallelization across HTCondor, SLURM, and PBS clusters, including NERSC; GPU-accelerated computing.
Programming	Python (NumPy, SciPy, Astropy, Pandas, JAX), C++, Mathematica.
Machine learning	PyTorch, TensorFlow, scikit-learn, gradient-boosted trees (XGBoost, CatBoost), Transformers, CNNs, distributed multi-GPU training.
Cosmology software & data products	SNANA, PIPPIN, SALT2/SALT3, TreeCorr, HEALPix.
Data handling	Large-scale FITS/HDF5 data handling.
Tools	Git/GitHub, Jupyter, LaTeX.

AWARDS & QUALIFICATIONS

- Qualified the TIFR Graduate School (GS) Admission Test and interview in Physics 2020
- Qualified the Joint Admission Test for M.Sc. (IIT JAM) in Physics 2020
- Shortlisted for interview at ICTS (via TIFR GS score) for Integrated Ph.D. in Physics 2020

SERVICE & OUTREACH

Organizing Committee Member, Young Astronomers’ Meet (YAM 2025) March 2025
TIFR, Mumbai | Meeting supported by the Astronomical Society of India for early-career researchers in astronomy, astrophysics, and cosmology.

Volunteer, Public Outreach Events
TIFR, Mumbai | Volunteered at TIFR public outreach events.

PROFESSIONAL MEMBERSHIPS

- Member, LSST Dark Energy Science Collaboration (DESC)